

# QARE-Watch: A Smartwatch-based Assistive System for Detecting the Quality of Arm Rehabilitation Exercises

Sirat Samyoun and John A. Stankovic

Department of Computer Science

University of Virginia

USA

{samyoun,stankovic}@virginia.edu

## ABSTRACT

Patients as well as healthy individuals often have to follow an exercise regimen at home that requires interactive support from a physical therapist. Although smartwatches have emerged as a great way for assisting daily exercises, existing systems do not focus on the quality factors of these exercises, such as, *missed repetitions, missed steps, and duration*. Towards addressing this notable gap in the literature, we present *QARE-Watch*, a smartwatch based system that efficiently measures these factors and assists the user accordingly. We conduct real life data collection from 12 participants and a preliminary evaluation in different settings to demonstrate the value of *QARE-Watch* for potential real life use.

## KEYWORDS

Arm exercises, Assistive Support, Smartwatch, Quality of exercise

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## 1 MOTIVATION

Patients with certain medical conditions (e.g., Stroke, Arthritis) often have to follow different arm based rehabilitation exercises for their functional recovery [2]. However, patients often fail to perform the exercises correctly without the help of a physical therapist, who typically monitors the quality of an exercise [2], and interactively guides the patient to do it correctly. Such in-home exercising are often recommended by health experts for the healthy individuals too. As such, we need intelligent assistive technologies for monitoring the quality of the exercises at home.

Modern smartwatches and fitness trackers have opened up tremendous possibilities to monitor user's daily physical activities in the wild. Several literature works [1, 6] as well as the smartwatch manufacturers have built several systems or apps for monitoring and supporting different basic daily activities, such as, walking, running, jogging, and sitting. However, none of these systems or apps

focused on detecting the quality factors of the performed exercises or on the exercises where such factors are crucial. Although several quality assessment and assistive systems have been proposed in the literature [3, 5], the quality factors and the assistance features varies for different domains. For example, a hand-hygiene quality monitoring system [3] usually focuses on whether every portion of the hand is properly cleaned following the standard steps, no matter which order the steps are performed. On the other hand, during a *shoulder flexion* arm exercise, a physical therapist visually checks if the user moves his/her affected arm from a resting position to the shoulder level, and then to straight above the head. The therapist warns the user if a step or a repetition is missed or not performed for enough duration. Moreover, all the steps have to be performed in the correct order. To accurately measure these factors, we need to capture the unique representation of each exercise step, and devise an algorithm for detecting the change points across the steps and repetitions, which has not been addressed in these literature works. Other quality assessment approaches in the literature used a video camera [1], which is highly privacy invasive and non-ubiquitous. Finally, none of the existing systems interactively guide the user based on the detected exercise quality, like a physical therapist. To solve the aforementioned challenges, we present *QARE-Watch*, a smartwatch based assistive system for the arm based exercises. *QARE-Watch* collects data from the user while performing an exercise, then utilizes a classifier and an algorithm for detecting the change points and the duration of the exercise steps, and guides the user to perform the exercise correctly.

## 2 SYSTEM OVERVIEW

*QARE-Watch* is a smartwatch based system designed for detecting the quality of arm based exercises and providing further assistance to the user. The functionalities of *QARE-Watch* are:

- (1) It provides support for different arm based exercises that involve movement of different parts of the arm. Such exercises are, *shoulder flexion, shoulder abduction, shoulder adduction, elbow flexion, and wrist flexion*.
- (2) Similar to a physical therapist, it assesses the quality factors of an exercise, such as, whether the user performed all steps of the exercise correctly and achieved the desired number of repetitions and duration. It then provides immediate feedback to the user about the quality of the exercise.
- (3) To overcome the small form factor limitation of the smartwatch, it interacts with the user by utilizing dialogue exchange capabilities. It also uploads the data to a cloud server enabling remote monitoring for the healthcare provider.

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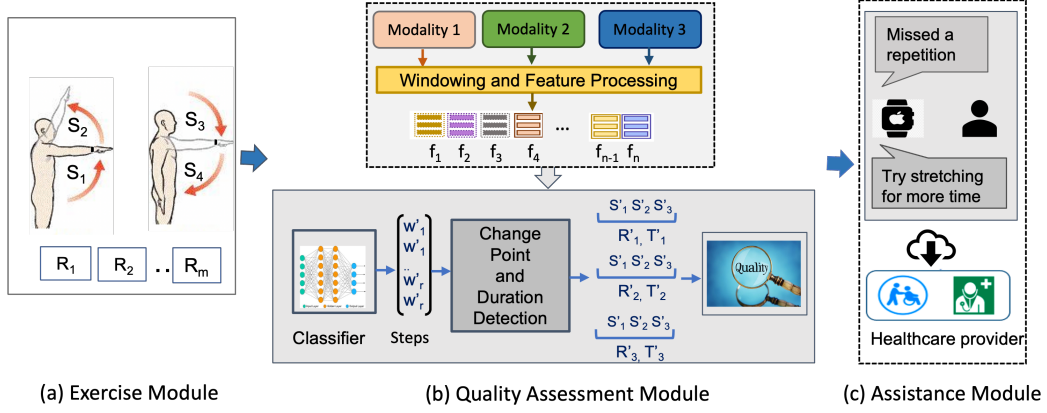


Figure 1: (a)-(c) Overview of the *QARE-Watch* architecture.

The system architecture of *QARE-Watch* is shown in Figure 1. It is divided into three key modules, which are discussed below.

**Exercise Module:** This module collects data from the user on the smartwatch while performing an exercise. An exercise consists of several independent steps ( $S_1, S_2, \dots, S_k$ ) that involves the movement for the wrist and elbow joint, as shown in Figure 1(a). A data signal is recorded from each motion sensing modality (e.g., accelerometer, gyroscope, magnetometer) available on the smartwatch at 50Hz sampling rate. Data is then split into small windows of size 0.06 seconds with 70% overlap between consecutive windows. These values are chosen based on the literature work [3], which ensures efficiently capturing the correlations within the sequential steps. Next, different statistical features such as, mean, standard deviation, kurtosis, and skew are extracted from each time window. The features are then scaled and combined to form the multimodal feature set ( $f_1, f_2, \dots, f_n$ ) for each window.

**Quality Assessment Module:** This module measures the quality factors of a performed exercise from the extracted multimodal features. The functionalities of this module are comprised of a classification part and a change point detection part.

**Classification:** This part applies a simple feature based machine learning classifier to the features and predicts a step of an exercise for each window. This allows making the model less complex and suitable for running on the resource constrained smartwatches compared to the deep models. The classifier receives the multimodal feature set of dimension  $n$  for each time window and outputs a single value that denotes a step of an exercise during that window. For all windows of count  $r$ , the predictions of the classifier are combined as  $W = (w'_1, w'_1, w'_1, \dots, w'_2, w'_2, w'_2, \dots, w'_r, w'_r)$ .

**Change point and duration detection:** This part runs an algorithm that identifies the start and end of a step and a repetition and then measures the respective duration(s) from the classification output. The sequential steps of the algorithm are as follows.

- First it iterates through the window set  $W$  and then finds the maximal continuous subset region of the windows in an incremental manner. Then it assigns a step value ( $S'_k$ ) to each subset. For example, a region having a maximum

occurrences of  $w'_1$  followed by multiple occurrences of  $w'_2$  is considered to be of the step  $S'_1$ .

- Next it identifies the probable region of a repetition by iterating through the assigned step values. Specifically, a repetition ( $R'_m$ ) is marked by the repeating values of ending step followed by those of the beginning step. At this point, the missed repetitions and missed steps are identified after running the algorithm over all data points.
- Finally, it measures the time duration of each performed step. This is calculated by using the pre-defined window length and the occurrences of the windows within each step region. Next, the average duration ( $T'_k$ ) of an exercise step is calculated across all repetitions. This helps to identify which step(s) were performed for insufficient duration.

**Assistance Module:** This module generates a feedback message based on the measured quality factors. The message contains which step(s) are missed by the user, if any. If the repetition count or the duration is insufficient compared to the desired count, this information is also included. Finally, it passes the message to the user by utilizing the text to speech framework of the smartwatch, as shown in Figure 1(c). It also uploads this information to a cloud server enabling remote monitoring for the healthcare providers (e.g., doctors, physical therapists).

### 3 USER STUDY AND EVALUATION

**Data Collection:** We conduct a real-life user study among 12 subjects consisting of the aforementioned five arm rehabilitation exercises. The participants were healthy individuals, aged between 26-39, including 7 male and 5 female persons. The participants wore the watch on one wrist and then performed 10 sessions of each exercise. It must be mentioned that most of the exercise rehabilitation programs focus on recovery on the affected arm, although both arms have to move in a similar manner. We utilized a data collection app presented in [4] that collects data from the 3 motion sensors, accelerometer, gyroscope, and magnetometer. The participants were shown instructional videos provided by a physical therapist for performing each exercise along with the details of

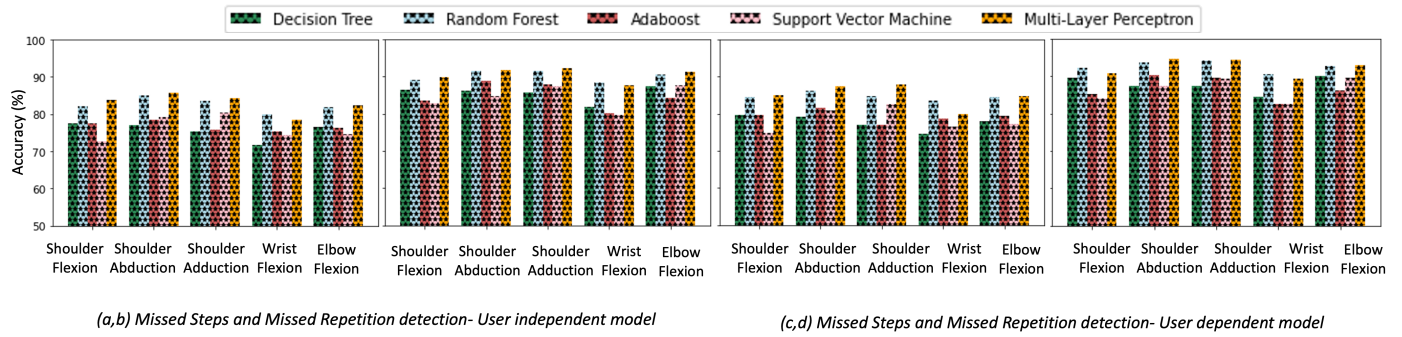


Figure 2: Preliminary evaluation results of QARE-Watch.

each step and the desired repetition count and duration. The sessions were video-recorded for ground-truth purpose. The average duration of exercising for each participant was 7 minutes.

**Experimental setup:** Following data collection, we split the data into training and testing sets, and trained a model for each exercise. To choose the best learning model, we experimented using different feature based machine learning classifiers. These are, *Decision Tree*, *Random Forest*, *Ada-boost*, *Support Vector Machine* and a *Multi-layered Perceptron* based neural network. We implemented the *Random Forest* classifier with 150 number of estimator trees, and with Gini impurity based information gain. The *Decision Tree* was also implemented using the Gini-based gains, and by following the best splitting strategy at each node. For the *Support Vector Machine* implementation, we used a classifier with the Radial Basis Function (RBF) kernel. The *Adaboost* implementation used 200 number of estimators and a learning rate of 0.01. The *Multi-layered Perceptron* implementation consisted of 3 layers and utilized the Relu activation function and the Adam optimizer. All these classifiers were implemented using the *Scikit-learn* library, and the aforementioned settings produced best results for our training environment. We measured the classification accuracy in two different settings:

- **User-independent model:** A model was trained using data from all participants except one, and the data for that one participant was used for testing. The procedure was repeated for each participant and the scores were averaged.
- **User-dependent model:** A model was trained for each participant using his/her data for all sessions except one, which was used for testing. The procedure was repeated for each session and the average scores were calculated.

**Results:** The average accuracy results for detecting the *missed repetitions* and *missed steps* in the user-independent and user-dependent settings are calculated and presented in Figure 2(a), 2(b), 2(c), 2(d) respectively. The average accuracy of detecting whether the *duration* was sufficient was found to be 99.9% across all exercises, as it is mostly dependent on the algorithm. We make the following observations from Figure 2.

- QARE-Watch is able to detect the *missed repetitions*, with an accuracy between 80.2%-92.3% and 82.6%-94.6% in the user-independent settings and the user-dependent settings,

respectively. In particular, the random forest and neural network based classifiers mostly provided the best results. The performance for *missed steps* detection is slightly lower than that of the *missed repetitions*.

- Measuring the quality factors of an exercise highly depends on the nature of the exercise. Some steps present in an exercise might involve similar kind of movements, which can hurt the performance of the model. In our experiments, *wrist flexion* yielded slightly lower performance than the other ones. Regardless, the overall results were very good.
- Our results also suggest that each individual has unique ways of performing an exercise, which can be better captured by the user-dependent or personalized model. That is why the user-dependent models yielded around 2%-3% better performance than that of the user-independent models.

## 4 CONCLUSION

QARE-Watch is designed to provide interactive assistance for the common arm exercises for the patients and healthy individuals. Future works will focus on studies demonstrating its capabilities among larger populations, handle the subjective differences for improving user personalization, and evaluating other quality metrics.

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